Brandon Yoo

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Education

McMaster University **Hamilton, ON, Canada**
Bachelor of Engineering & Biomedical Engineering, B.Eng.BME. | Bachelor of Health Sciences, B.H.Sc. *Sep 2024 – May 2029*
Majors: Chemical & Biomedical Engineering, Honours Biochemistry.

Scholarships & Awards [selected]

- **Cansbridge Fellowship ($10k),** 1/17 scholar recipients sponsored to pursue an internship experience in Asia.
- **McMaster Dean's Honours List 2026,** awarded to students who maintain above a 9.5/12.0 GPA for the 2025-2026 academic year.
- **Professor Terrance Hoffman Scholarship ($6k),** awarded to the top student entering the Department of Chemical Engineering.
- **Elizabeth Jenkins Scholarship ($6.5k).**
- **McMaster Provost's Honours List 2025,** awarded to students who maintain a 12.0/12.0 GPA for the 2024-2025.
- **McMaster Dean's Excellence Award ($7.5k).**
- **McMaster Award of Excellence ($3k).**

Professional Experience

Samsung Medical Center – 지니너스 **Seoul, KR**
AI R&D Intern *May 2026 – Present*
- Developing a sequential multi-model pipeline for breast cancer drug target identification, leveraging in-house spatial transcriptomics data not available in public datasets.
- Applying diffusion models for denoising and imputing of high-dimensional spatial transcriptomics data, feeding learned representations into a Graph Neural Network (GNN) for downstream target scoring.
- Prototyping ML/DL architecture integrating tissue-level spatial gene expression context into the target identification workflow.

University of Calgary – Center for Health Informatics **Calgary, AB, CA**
Research Associate *Apr 2025 – Present*
- Prototyping a Neural ODE-based triage model for real-time stroke severity scoring across disposition categories (ICU, general ward, discharge), leveraging continuous-time modeling of patient trajectories from heterogeneous EMR data including vitals, labs, clinical notes, imaging, medications, and admission records.
- Collaborating with the UHN Krembil Brain Institute on model development, utilizing real-world clinical EMR data sourced directly from UHN's neurology patient cohort.

McMaster Immunology Research Center – Rullo Lab **Hamilton, ON, CA**
Research Assistant *Nov 2025 – Present*
- Sole dry-lab researcher embedded in a wet-lab immunology team, providing end-to-end computational drug optimization support for cancer immunology research.
- Optimizing covalent ligand binding affinity against the CD206 receptor through iterative electrophile positioning guided by binding pocket geometry, cysteine proximity, and molecular docking scores.
- Building and executing cheminformatics pipelines and MD simulations to evaluate ligand conformations and inform experimental synthesis.

Curnew Medicine **Hamilton, ON, CA**
Research Coordinator *Jan 2025 – Apr 2026*
- Conducted initial patient evaluations for 10+ patients per week in a cardiovascular clinic, independently performing vitals, medical history intake, symptom review, and interpretation of diagnostic tests (ECG, echocardiogram, Holter monitor, stress test) prior to consultation.
- Accumulated 150+ hours of direct patient contact, developing clinical fluency in cardiovascular presentation and diagnostic workflows.
- Assisted in the coordination of ongoing clinical trials, spanning patient recruitment and screening, informed consent, data collection, adverse event tracking, and sponsor communication.

Myant Corporation **Mississauga, ON, CA**
Data Science Intern *May 2025 – Aug 2025*
- Built a centralized analytics dashboard in Python, SQL, and Excel directly commissioned by and reported to the COO, enabling company-wide visibility into SKIIN device usage and performance metrics for investor updates and internal reviews.
- Quantified failure rates across device batches by analyzing device logs and adherence data, isolating connectivity dropout, sensor signal quality, and battery issues to specific firmware, hardware, and garment design changes.
- Dashboard remained in active company-wide use post-internship, serving as the primary tool for device performance tracking.

Leadership Experience

McMaster Varsity Dance Team **Hamilton, ON, CA**
Captain *Apr 2026 – Present*
- Co-leading a 30-person competitive varsity dance team, overseeing choreography, training, auditions, scheduling, and fundraising.

Nucleate Dojo **Hamilton, ON, CA**
Program Manager – McMaster Chapter *Jan 2025 – Present*
- Direct operations for the McMaster chapter of a Harvard-founded biotech entrepreneurship nonprofit, leading programming and events that attract 50–100 attendees per event.

Projects [selected]

YooArt – Enterprise Performance Planning Software [Link] *May 2026 – Present*
- Building a desktop application that extracts full-body + finger-level motion data from choreography video and pre-visualizes them in an open 3D production environment.

Directed Evolution of the cspA Cold-Shock Regulatory Element [Link] *Apr 2025 – Oct 2025*
- Applied RNA thermodynamics modeling (RBS Calculator) to predict and rank 5'UTR sequence variants of the *E. coli* cspA cold-shock element, identifying mutations with up to 2.6x increase in translation initiation rate for cold-inducible recombinant protein expression.

Skills

Technical: Programming (Python, R, SQL, JavaScript/TypeScript); FOSS & Development Tools (Git, Vim, Linux, Bash, LaTeX); Machine Learning (PyTorch, TensorFlow, JAX, scikit-learn, OpenCV); Bioinformatics & Computational Biology (bulk RNA-seq, scRNA-seq, CITE-seq, spatial omics, multi-modal data integration, ligand-receptor inference, molecular docking, molecular dynamics).

Laboratory: Molecular Cloning (Miniprep/Maxiprep, DNA ligation, bacterial transformation); PCR & Gel Electrophoresis; Bacterial & Mammalian Cell Culture; Flow Cytometry; Mass Spectrometry; IR, UV-Vis & NMR Spectroscopy; Chromatography.